

KOLMARDEN-STROMSFORS, SwedenWMO: **025580**Lat: **58.689N**Lon: **16.307E**Elev: **155**StdP: **99.48**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.4	-11.8	-16.7	0.9	-12.7	-14.5	1.1	-9.7	8.4	4.2	7.4	3.4	3.0	340	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	27.0	17.6	25.0	17.0	23.2	16.2	19.2	24.0	18.2	22.6	17.2	21.6	2.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.9	20.5	16.6	12.0	19.6	15.6	11.3	18.7	55.4	24.4	51.9	22.6	49.0	21.6	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.6	5.1	-15.9	30.3	3.1	2.4	-18.1	32.0	-20.0	33.4	-21.7	34.8	-24.0	36.5		
			-16.2	20.9	3.0	1.4	-18.4	21.9	-20.1	22.7	-21.8	23.4	-24.0	24.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.5	-2.1	1.0	6.0	11.3	15.4	17.8	16.8	12.3	6.8	2.4	-0.9
	DBStd	8.09	4.44	4.30	4.12	3.67	3.56	3.11	3.04	2.64	2.95	3.44	3.60	4.56
	HDD10.0	1847	388	339	279	128	26	1	0	0	12	108	228	336
	HDD18.3	4164	647	572	537	369	220	99	47	63	181	357	478	594
	CDD10.0	782	0	0	0	9	65	162	243	211	82	9	0	0
	CDD18.3	58	0	0	0	0	1	9	32	15	1	0	0	0
	CDH23.3	391	0	0	0	0	14	83	218	74	1	0	0	0
	CDH26.7	69	0	0	0	0	1	14	44	11	0	0	0	0
Wind	WSAvg	2.7	3.0	2.9	2.9	2.6	2.5	2.6	2.5	2.5	2.7	2.8	2.9	2.9
Precipitation	PrecAvg	609	38	36	28	35	42	65	75	72	58	53	57	48
	PrecMax	785	72	63	54	84	79	110	139	162	137	139	124	93
	PrecMin	507	5	14	7	3	9	11	4	9	9	16	11	12
	PrecStd	60	18	12	13	23	18	25	34	36	36	29	28	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	8.3	14.2	19.7	25.1	28.6	30.2	27.8	22.7	15.8	11.1	9.1
		MCWB	6.1	5.7	8.0	11.7	14.5	16.8	19.0	18.7	16.1	12.8	9.6	7.8
	2%	DB	5.5	6.1	10.9	17.0	22.4	25.7	27.8	25.3	20.5	13.9	9.6	6.9
		MCWB	4.2	4.1	6.2	9.9	13.6	16.5	18.1	17.8	15.2	11.9	8.6	5.8
	5%	DB	4.2	4.7	8.5	14.4	20.0	23.4	25.7	23.3	18.5	12.7	8.3	5.7
		MCWB	3.2	3.4	5.0	8.7	12.7	15.3	17.5	16.9	14.1	11.1	7.5	4.9
	10%	DB	3.0	3.5	6.5	12.1	17.7	21.4	23.8	21.6	16.8	11.5	7.1	4.5
		MCWB	2.3	2.4	3.9	7.1	11.4	14.2	16.9	16.1	13.3	10.2	6.4	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	6.6	8.6	12.5	16.5	19.0	20.8	20.9	17.2	13.7	10.4	8.0
		MCDB	7.2	7.7	13.2	18.3	21.5	23.9	26.3	25.1	20.6	15.0	10.7	9.1
	2%	WB	4.5	4.7	6.7	10.5	14.9	17.3	19.5	19.0	15.8	12.4	8.8	6.1
		MCDB	5.2	5.6	10.2	15.6	19.8	23.1	24.6	23.1	19.0	13.5	9.3	6.8
	5%	WB	3.3	3.6	5.5	9.0	13.4	16.2	18.6	17.9	14.8	11.4	7.6	4.9
		MCDB	4.0	4.7	7.9	13.7	18.2	21.9	23.6	21.3	17.6	12.4	8.1	5.5
	10%	WB	2.3	2.4	4.2	7.6	12.2	15.1	17.6	17.0	13.8	10.4	6.6	3.9
		MCDB	2.9	3.5	6.1	11.4	16.8	20.1	22.4	20.5	16.2	11.3	7.0	4.4
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	6.6	8.8	9.5	9.4	9.0	8.3	7.2	5.3	3.7	3.8
		MCDBR	4.8	5.4	9.3	11.9	12.6	12.4	11.6	10.3	9.1	6.2	5.0	4.7
		MCWBR	4.6	4.1	5.8	6.6	6.3	5.6	4.5	4.5	4.8	4.3	4.6	4.5
	5% WB	MCDBR	4.7	5.1	8.7	11.2	11.2	11.1	10.2	9.0	8.3	5.7	4.6	4.4
		MCWBR	4.7	4.1	5.7	6.6	6.3	5.7	4.7	4.6	4.9	4.5	4.3	4.4
Clear-Sky Solar Irradiance	taub	0.278	0.315	0.333	0.370	0.366	0.355	0.381	0.373	0.341	0.332	0.302	0.263	
	taud	2.249	2.312	2.371	2.332	2.406	2.458	2.403	2.424	2.499	2.434	2.293	2.174	
	Ebn at Noon	553	709	811	836	866	881	845	826	801	687	513	431	
	Edh at Noon	45	69	86	106	105	101	105	96	76	61	43	34	
All-Sky Solar Radiation	RadAvg	0.33	0.93	2.43	3.97	5.24	5.80	5.29	4.21	2.81	1.26	0.43	0.23	
	RadStd	0.05	0.16	0.25	0.40	0.43	0.28	0.51	0.40	0.34	0.20	0.09	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-129	-158	N/A

Nomenclature: See separate page